

Boolean Algebra

Variables & operations · all the laws · De Morgan · consensus · duality · simplifying step by step

QUICK TRICK

Green boxes = quick tricks. Boolean algebra looks scary, but there are only about 10 rules. Learn the tricks and simplification becomes a game.

4.1 What is Boolean algebra?

● LEVEL 1 · BEGINNER

What is it? Normal algebra works with many numbers. **Boolean algebra works with only two values: 0 and 1.** We give a value a name like A, B or C. These names are called **variables**. Numbers 0 and 1 are called **constants**.

Why do we need it? Every digital circuit is a Boolean expression in disguise. If we can make the expression shorter, the circuit needs fewer gates. Fewer gates = cheaper, faster, less power.

Meaning of 0 and 1: 0 = FALSE = OFF = LOW. 1 = TRUE = ON = HIGH.

An **expression** joins variables using three operations: AND, OR and NOT.

Operation	Symbol	Simple meaning	Real-life example
AND	$A \cdot B$ or AB	Answer is 1 only if both are 1	To enter the exam hall you need admit card AND ID card
OR	$A + B$	Answer is 1 if at least one is 1	You can pay by cash OR card
NOT	A' (say "A bar")	Flips the value: $0 \rightarrow 1$, $1 \rightarrow 0$	The opposite of ON is OFF

A	B	$A \cdot B$ (AND)	$A + B$ (OR)	A' (NOT A)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

EASY MEMORY

AND is like multiplication (anything times 0 is 0). **OR is like addition, but $1 + 1 = 1$** (not 2!). Only 0 and 1 exist.

QUICK TRICK

Order of work: N-A-O = NOT first, then AND, then OR. So $A + B \cdot C'$ means $A + (B \cdot (C'))$. Use brackets to change the order.

4.2 The laws of Boolean algebra

● LEVEL 2 · EXAM

A **law** is a rule that is always true. Use these rules to make expressions shorter. Below, every law is written in simple words first.

Law	OR form	AND form	In simple words
Identity	$A + 0 = A$	$A \cdot 1 = A$	Adding a 0 (or multiplying by 1) changes nothing
Null (Dominance)	$A + 1 = 1$	$A \cdot 0 = 0$	A 1 in an OR wins. A 0 in an AND wins
Idempotent	$A + A = A$	$A \cdot A = A$	Repeating a variable does nothing
Complement	$A + A' = 1$	$A \cdot A' = 0$	A value and its opposite: OR gives 1, AND gives 0
Involution	$(A')' = A$		Two NOTs cancel each other
Commutative	$A + B = B + A$	$AB = BA$	Order does not matter
Associative	$(A+B)+C = A+(B+C)$	$(AB)C = A(BC)$	Grouping does not matter
Distributive	$A(B + C) = AB + AC$	$A + BC = (A+B)(A+C)$	The second form is special: it does NOT work in normal algebra
Absorption	$A + AB = A$	$A(A + B) = A$	The small term swallows the big term
Redundancy (extra)	$A + A'B = A + B$	$A(A' + B) = AB$	A complemented variable can be removed
De Morgan	$(A + B)' = A' B'$	$(AB)' = A' + B'$	Break the bar and change the sign
Consensus	$AB + A'C + BC = AB + A'C$	$(A+B)(A'+C)(B+C) = (A+B)(A'+C)$	The “consensus” term BC is not needed

COMMON MISTAKE

Treating Boolean algebra like normal algebra. Remember: $A + A = \mathbf{A}$ (not $2A$), $A \cdot A = \mathbf{A}$ (not A^2), $A + 1 = \mathbf{1}$.

Truth table proofs (so you trust the laws)

If two expressions have the **same output on every row**, they are equal. Compare the last two columns.

Absorption: $A + AB = A$

A	B	AB	A + AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1

Redundancy: $A + A'B = A + B$

A	B	A'B	A + A'B	A + B
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1
1	1	0	1	1

Column “A + A'B” and column “A + B” are identical, so the law is true.

De Morgan: $(AB)' = A' + B'$ and $(A + B)' = A'B'$

A	B	(AB)'	A'+B'	(A+B)'	A'B'
0	0	1	1	1	1
0	1	1	1	0	0
1	0	1	1	0	0
1	1	0	0	0	0

Second distributive law: $A + BC = (A + B)(A + C)$

A	B	C	BC	A + BC	(A+B)(A+C)
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	1	1	1

Consensus: $AB + A'C + BC = AB + A'C$

A	B	C	AB	A'C	BC	AB+A'C+BC	AB+A'C
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	1
0	1	0	0	0	0	0	0
0	1	1	0	1	1	1	1
1	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0
1	1	0	1	0	0	1	1
1	1	1	1	0	1	1	1

4.3 Tricks to remember the big laws

De Morgan's law

QUICK TRICK

“Break the bar, change the sign.”

$(A \cdot B)'$ → break the bar into A' and B', change AND to OR → **$A' + B'$**

$(A + B)'$ → break the bar into A' and B', change OR to AND → **$A' \cdot B'$**

For 3 or more variables it works the same: $(A + B + C)' = A' B' C'$ and $(ABC)' = A' + B' + C'$.

QUICK TRICK

Complement of a big expression — fast method. Swap every + with · (and · with +), and put a bar on every variable (remove the bar if it already has one).

Example: $(A + B'C)' \rightarrow A' \cdot (B + C') = \mathbf{A'(B + C')}$.

Example: $(AB' + C'D)' \rightarrow (A' + B)(C + D')$.

COMMON MISTAKE

$(A + B)'$ is **NOT** $A' + B'$. It is **$A'B'$** . The sign must change when the bar is broken.

Absorption

QUICK TRICK

“The small term swallows the big term.” If a short term (like A) appears, any longer term that *contains* it (like AB, ABC, ABD...) is not needed. $A + AB = A$. $A + ABC = A$. For the AND form: $A(A + B) = A$.

Redundancy (the most useful trick)

QUICK TRICK

“ $A + A'B = A + B$.” When you see a variable and, in another OR term, the **opposite** of that variable multiplied by something, the opposite variable disappears.

Examples: $B + B'C = B + C$ | $X + X'Y = X + Y$ | $A + A'BC = A + BC$.

AND form: $A(A' + B) = AB$.

Consensus

QUICK TRICK

Consensus trick. Look for two terms where one has a variable A and the other has A' (like AB and A'C). Join the **other** parts (B and C) into the term **BC**. If BC is present in the expression, **cross it out**. It is not needed.

Example: $AB + A'C + BC = AB + A'C$. Also BCD (bigger than BC) is not needed either.

AND form: $(A + B)(A' + C)(B + C) \rightarrow (A + B)(A' + C)$.

Second distributive law

QUICK TRICK

$A + BC = (A + B)(A + C)$. Think: “A can go inside both brackets.”

Use it to change a sum of products into a product of sums. Example: $A + B'C = (A + B')(A + C)$.

4.4 Duality principle

● LEVEL 2 • EXAM

Dual of an expression: swap **AND** $\leftrightarrow$ **OR** and swap **0** $\leftrightarrow$ **1**. Keep the variables as they are. **Rule:** if a law is true, its dual is also true. That is why laws come in pairs (OR form and AND form).

Expression	Dual (swap + and $\cdot$, 0 and 1)	Complement (different thing!)
$A + B$	$A \cdot B$	$A' B'$
$A(B + C')$	$A + B C'$	$A' + B' C$
$A B + 0$	$(A + B) \cdot 1$	$(A' + B') \cdot 1$
$A + 1 = 1$	$A \cdot 0 = 0$ (also true)	—

QUICK TRICK

Dual = swap the operators and the 0/1, but keep the variables. **Complement** = also change the variables. Do not mix them up.

4.5 Simplifying expressions step by step

● LEVEL 2 • EXAM

HOW TO SIMPLIFY (follow this order)

- 1) Remove brackets and NOT-bars over big parts using De Morgan.
- 2) Look for a COMMON FACTOR (like A) in two terms $\rightarrow$ take it out.
- 3) Look for $X + X' = 1$ or $X \cdot X' = 0$.
- 4) Look for ABSORPTION ($A + AB = A$).
- 5) Look for $A + A'B = A + B$.
- 6) Look for CONSENSUS.
- 7) Stop when none of the rules can be used.

QUICK TRICK

Lazy check trick: after simplifying, put $A = 0$ and $A = 1$ (and the same for B, C) into both the original and the final expression. If the answers match for every combination, you are correct. (Or compare truth tables.)

Example 1 (easy): $AB + AB'$

Take the common factor A: $A(B + B')$

$B + B' = 1$, so $A \cdot 1 = \mathbf{A}$

Example 2: $A + A'B + AB$

$A + A'B = A + B$ (redundancy trick). Now we have $A + B + AB$.

AB is swallowed by A (absorption). Answer: $\mathbf{A + B}$

Example 3: $(A + B)(A + C)$

Multiply: $AA + AC + BA + BC = A + AC + AB + BC$ (since $AA = A$)

A swallows AC and AB . Answer: $\mathbf{A + BC}$ (the second distributive law)

Example 4: $A'B'C + A'BC + AB'C + ABC$

Every term has C. Take C out: $C(A'B' + A'B + AB' + AB)$

Inside the bracket all four combinations of A and B appear, so it equals 1. Answer: $\mathbf{C}$

Trick: if all combinations of some variables appear together, those variables disappear.

Example 5 (De Morgan): $((A + B)' + C)'$

Break the outside bar: $((A + B)')' \cdot C' \rightarrow (A + B) \cdot C'$

Answer: **$(A + B) C'$**

Example 6 (consensus): $AB + A'C + BC$

B and C are the “other parts” of AB and A'C, and BC is present → cross it out.

Answer: **$AB + A'C$**

Example 7 (consensus, AND form): $(A + B)(A' + C)(B + C)$

Same idea: B + C is the consensus of (A + B) and (A' + C). Answer: **$(A + B)(A' + C)$**

Example 8 (Level 3): $A'B + AB' + AB$

$A'B + AB = B(A' + A) = B$. Now we have $B + AB'$.

$B + AB' = A + B$ (redundancy trick with the roles swapped: $B + B'A = B + A$).

Answer: **$A + B$**

Example 9 (Level 3): $(A + B)(A' + B)(A + B')$

$(A + B)(A' + B) = B + AA' = B + 0 = B$ (second distributive law).

$B \cdot (A + B') = AB + BB' = AB + 0 = \mathbf{AB}$

4.6 Common mistakes and exam traps

COMMON MISTAKE

- $(A + B)'$ written as $A' + B'$ (wrong). Correct: $A'B'$.
- $A + A = 2A$ (wrong). Correct: A .
- $(AB)'$ written as $A'B'$ (wrong). Correct: $A' + B'$.
- Thinking $A + BC = (A + B) + C$. It is $(A + B)(A + C)$.
- Forgetting a bar when you apply De Morgan.

EXAM TRAP

GATE often asks: “which of these expressions is equal to F?”. Do not simplify everything. **Test with numbers**: put A, B, C = 0 or 1 and check which option matches. It is faster.

GATE POINT

Consensus theorem and **De Morgan** are the two most tested laws. Learn both forms (OR form and AND form).

Quick-Solve Tricks — Summary

Situation	Trick
See a bar over a big part	Break the bar, change the sign (AND $\leftrightarrow$ OR), bar every variable.
$X + X \cdot (\text{something})$	Absorption: the short term swallows the long term.
$X + X' \cdot Y$	Becomes $X + Y$. The X' disappears.
$X \cdot (X' + Y)$	Becomes $X \cdot Y$.
$AB + A'C (+ BC)$	BC is the consensus. Cross it out.

Situation	Trick
All combinations of a variable together	That variable disappears ($A + A' = 1$).
Two terms differ in one variable	Factor and remove that variable: $AB + AB' = A$.
Need the dual	Swap $+$ $\leftrightarrow$ $\cdot$ and $0 \leftrightarrow 1$. Keep variables.
Not sure if two expressions are equal	Try all 0/1 values (truth table).
Order of operations	N-A-O: NOT, then AND, then OR.

GATE QUICK CHECK — Chapter 4

Identity : $A + 0 = A$	$A \cdot 1 = A$	Null : $A + 1 = 1$	$A \cdot 0 = 0$
Idempotent: $A + A = A$	$A \cdot A = A$	Complement: $A + A' = 1$	$A \cdot A' = 0$
Absorption: $A + AB = A$	$A(A + B) = A$	Redundancy: $A + A'B = A + B$	
De Morgan : $(A+B)' = A'B'$	$(AB)' = A' + B'$	Distributive: $A + BC = (A+B)(A+C)$	
Consensus : $AB + A'C + BC = AB + A'C$		$(A+B)(A'+C)(B+C) = (A+B)(A'+C)$	

Traps: De Morgan sign change $\cdot$ dual is not complement $\cdot 1 + 1 = 1$ $\cdot$ second distributive law $\cdot$ consensus term may be hidden inside a bigger term.

Practice questions

● LEVEL 1 • BEGINNER

Q1. Simplify $(A + 1) \cdot B$.

Q2. What is $(A')' + A \cdot 1$?

Q3. Write the dual of $A + B \cdot 1$.

● LEVEL 2 • EXAM

Q4. Simplify $AB + AB'C$.

Q5. Find the complement of $A + B'C$.

Q6. Simplify $A'B + A'B' + AB'$.

● LEVEL 3 • GATE

Q7. Simplify $AB + A'C + BCD$.

Q8. Simplify $(A + C)(A' + B)(B + C)$.

Q9. True or false: $A + BC = (A + B)(A + C)$.

Q10. True or false: $(A + B)' = A' + B'$.

Q11. Simplify $(A + B + C)(A + B + C')(A + B' + C)$.

Q12. How many literals are in the simplest form of $A'B + AB' + AB$? (A literal is one variable, with or without a bar.)

Answers

Q	Solution
1	$A + 1 = 1$ (null law). $1 \cdot B = \mathbf{B}$.
2	$(A')' = A$ (involution), $A \cdot 1 = A$. $A + A = \mathbf{A}$.
3	Swap $+$ and $\cdot$, and 1 becomes 0: $\mathbf{A \cdot (B + 0)}$.

Q	Solution
4	Take A out: $A(B + B'C)$. Redundancy trick: $B + B'C = B + C$. Answer $A(B + C) = AB + AC$.
5	Fast method: swap operators and bar the variables. $A' \cdot (B + C')$. Answer $A'(B + C')$. (Check with De Morgan: $(A + B'C)' = A' \cdot (B'C)' = A'(B + C)$.)
6	$A'B + A'B' = A'(B + B') = A'$. Then $A' + AB' = A' + B'$ (redundancy). Answer $A' + B'$.
7	A and A' are in AB and A'C. Consensus is BC. BCD contains BC, so it is not needed. Answer $AB + A'C$.
8	Rename: $(A + C)(A' + B)(C + B)$ has the shape $(A + X)(A' + Y)(X + Y)$ with $X = C$, $Y = B$. The last bracket is the consensus. Answer $(A + C)(A' + B)$.
9	True (second distributive law; proof by truth table above).
10	False. Correct: $(A + B)' = A'B'$.
11	$(A + B + C)(A + B + C') = A + B$ (the C variable disappears: $(A+B) + CC'$). Then $(A + B)(A + B' + C) = A + B(B' + C) = A + BC$. Answer $A + BC$. (Check with $A=0, B=1, C=0$: the third bracket $A + B' + C = 0 + 0 + 0 = 0$, so the original is 0. And $A + BC = 0 + 0 = 0$. ✓)
12	$A'B + AB = B$. $B + AB' = A + B$. Answer $A + B$ has 2 literals .

Next: Chapter 5 — Logic Gates (AND, OR, NOT, NAND, NOR, XOR, XNOR, buffer, universal gates, and quick tricks for building circuits).